

EAN CUMMINGS

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EDUCATION

University of Iowa – College of Engineering

Iowa City, IA

Master of Science in Electrical and Computer Engineering (M.S.)

Anticipated: December 2026

- **Focus:** Integrated Sensing & Critical Infrastructure
- **GPA:** 3.66

Bachelor of Science in Biomedical Engineering (B.S.E.)

May 2025

- **Focus Area:** Bioimaging; Image and Signal Processing, Embedded Systems
- **Study Abroad:** Dublin, Ireland (STEM & Irish Studies)

TECHNICAL SKILLS

- **Programming Languages:** Swift, Python, C++, Java, MATLAB, R, Alloy, Dafny
- **Hardware & Cleanroom Fabrication:** Photolithography, RF Magnetron Sputtering, Wet/Dry Etching, VASE, SEM, Circuit Design, Microcontrollers (ESP32/Arduino), Oscilloscopes, Precision TIG Welding (Titanium/Steel)
- **Software, Tools & Frameworks:** AutoCAD, Creo Parametric, ImageJ, Git, Firebase, InfluxDB, PSCAD, Power World

EXPERIENCE

LiDAR Systems Laboratory (NASA-Sponsored) | University of Iowa

Jan 2026 – May 2026

- Modeled and calibrated beam-steering optics for atmospheric research, validating range-dependent signal scaling and detector-lens FWHM response.
- Characterized photodiode transient responses and optimized TIA feedback compensation networks to eliminate capacitance-induced oscillations.

Microfabrication & Thin Film Laboratory | University of Iowa

Jan 2026 – May 2026

- Lead end-to-end device fabrication flows, including bilayer lithography, reactive RF magnetron sputtering, and anisotropic KOH etching.
- Characterized film uniformity and sidewall geometry using VASE Cauchy modeling and SEM imaging to ensure stoichiometric growth.

PROJECTS

PuppyPOV – Wearable IoT Pet Health Monitor

- Designed an integrated smart harness to track canine biometrics (Heart Rate, HRV, SpO2 and motion) and environmental context using a Raspberry Pi hub, PiCam, and I2C sensors.
- Engineered a custom DSP pipeline, utilizing bandpass filtering and peak detection to extract vital signs from motion-corrupted signals and stream data to Firebase/InfluxDB.

InsulTrack – Smart Insulin Pen Case & iOS App

- **Award:** 3rd Place, Hubert E. Storer Engineering Student Entrepreneurial Start-Up Award.

- Designed a thermally insulated, sensor-embedded smart case for diabetic management featuring dosage logging, GPS tracking, and environmental protection.
- Engineered a companion native iOS application using Swift and CoreBluetooth (BLE) for real-time data synchronization of blood glucose, insulin usage, exercise, and nutritional metrics.

Capsule Closet – Generative-AI Wardrobe Web Application

- Developed a web app that parses uploaded clothing photography to generate AI outfit try-on models via the Gemini API.

Electrocardiogram (EKG) Design & Build

- Constructed a functional 3-lead EKG using multi-stage instrumentation amplifiers and analog filtering circuits.
- Interfaced hardware with an Arduino microcontroller for real-time heart rate plotting and automated atrial fibrillation (AFib) detection.

Custom Titanium and Steel Fabrication

- Engineered custom bicycle frames using AutoCAD for geometry optimization, tube-set profiling, and frame-jig alignment.
- Executed structural TIG welding on titanium and steel alloy elements, demonstrating strict atmospheric and material contamination control.

LEADERSHIP

Lead Student Ambassador

University of Iowa College of Engineering | 2024 – 2025

- Trained, scheduled, and mentored a team of 18 student ambassadors, directing campus tours, information sessions, and panel discussions for prospective students and families.
- Organized and executed large-scale collegiate recruitment events and coordinated academic expert panels.

Iowa Center for Advancement Ambassador

University of Iowa | 2022-2023

- Advocated for University of Iowa by highlighting students achievements to university donors and alumni.
- Collaborated with university administration to execute high-profile institutional events and executive hosting initiatives for the Office of the President.

President

University of Iowa Triathlon Team | 2024 – 2025

- Manage team operating budgets, corporate sponsorships, travel logistics, and weekly training schedules for club members.
- Organized and hosted Triowa (the University of Iowa indoor triathlon event) and competed at the USAT Collegiate Club National Championships.